

# Zhi Zhou

(+358) 045 325 6823

[zhi.zhou@helsinki.fi](mailto:zhi.zhou@helsinki.fi)

## PROFILE

---

Master's student in Bioinformatics and Systems Medicine at the University of Helsinki (in progress), with a background in statistics and applied health and population research. Skilled in statistical and machine-learning modelling, longitudinal and survival analysis, and reproducible research using R, SQL, and Python. Looking for an MSc thesis opportunity in bioinformatics and precision medicine, particularly in modelling patient responses and disease progression.

## EDUCATION

---

**University of Helsinki** – Helsinki, Finland 2026.08-present

Master of Science in Life Science Informatics

- Focus areas: Bioinformatics and Systems Medicine

**Stockholm University** – Stockholm, Sweden 2016.09-2021.02

Master of Science in Statistics

- Relevant coursework: Bayesian Statistics; Statistical Computation; GLMs; Multivariate Analysis; Experimental Design
- Master's thesis: "The Effect of Auxiliary Variable(s) in Imputation with Logistic Regression" (Supervisor: Per Gösta Andersson).

## PROFESSIONAL EXPERIENCE

---

**University of Helsinki** – Helsinki, Finland 2025.11-present

Biostatistician (contract)

- Assist researchers in planning, implementing, analysing, interpreting, reporting, and publishing medical research projects.
- Contribute to the design and delivery of biostatistics and statistical software (SPSS and R) training for staff.

**Finnish Institute for Health and Welfare** – Helsinki, Finland 2020.08-2024.12

Statistical Researcher

- Designed and implemented statistical analyses for national health and population studies using R (and SQL as needed).
- Developed R packages for simulation and projection modelling (including multiple-imputation-based workflows) with comprehensive user guides.
- Applied regression and survival models (e.g., GLMs, Cox models, longitudinal analyses) and communicated results to both statisticians and domain experts.
- Co-authored peer-reviewed publications in multidisciplinary research teams.

## RESEARCH EXPERIENCE

---

**Projections for effects of changes in non-communicable diseases and their risk factors on the needs and costs of health services** 2023.01-2024.12

- The work was carried out with funding received by THL from the Sustainable Growth Programme under the EU's one-off recovery instrument (Next Generation EU).
- Conducted analyses employing multiple imputation, Cox regression, and illness-death models.
- Modelled service needs using multimorbidity indicators at the regional level; produced publication-ready figures and summaries.
- Implemented reusable R functions and a user guide for end users.

**The use of performance measures based on the client's self-assessment supports the planning and evaluation of rehabilitation** 2022.08-2023.11

- Extracted and quality-checked data from the Lomakepalvelu (LOPA) database and published summary outputs using R.
- Contributed to the preparation of the final report and supported colleagues with analyses and figures.

**Addressing the smoking paradox in the etiology of COVID-19 through population-based studies (Tobrisk-Cov)** 2022.09-2023.09

- Cleaned the vaccine data, ensuring that the vaccine names and batch numbers were accurately reflected in the records.
- Cross-checked each entry against the original sources to guarantee the accuracy and precision of the information.

- Conducted a comprehensive series of analyses to achieve the research goals. This included stratified and sensitivity analyses, as well as an explanatory survival analysis.

### Disability trends and projections in Finland

2022.03-2022.10

- This work is conducted as part of Finland's Recovery and Resilience Plan (RRP), a component of the Sustainable Growth Programme for Finland and has received funding from the EU Recovery Plan 'Next Generation EU' and the Recovery and Resilience Facility (RRF).
- Performed descriptive analyses of the data to gain an in-depth understanding of underlying patterns, trends, and relationships.
- Conducted an analysis of work ability by birth cohort.
- Provided analytical and organisational support to colleagues in preparing and delivering conferences.

### Projections of the burden of disease and disability in Finland – health policy respects (PoDDy-HePo)

2020.08-2021.12

- Harmonised data and projected future trends in current smoking, sedentary lifestyle, hypertension, elevated total cholesterol, BMI categories, and diabetes.
- Developed an R package that includes functions for model selection, projection, pooling, and plotting, all built on the principles of multiple imputation (MI). Additionally, compiled a comprehensive user guide to facilitate seamless utilisation of the package.

## SELECT PUBLICATIONS

---

- Lahti, J., Reinikainen, J., Kontto, J., **Zhou, Z.**, Koskinen, S., Laaksonen, M., Partonen, T., Elonheimo, H., Lundqvist, A., & Tolonen, H. (2024). Work ability trends 2000-2020 and birth-cohort projections until 2040 in Finland. *Scandinavian journal of public health*, 14034948241228155. Advance online publication. <https://doi.org/10.1177/14034948241228155>
- Peña, S., **Zhou, Z.**, Kestilä, L., Galanti, M. R., Shaaban, A. N., Caspersen, I. H., Magnus, P., Geraldo, P., Rojas-Saunero, P., Parikka, S., Nohynek, H., & Karvonen, S. (2024). Tobacco Use and Uptake of COVID-19 Vaccinations in Finland: A Population-Based Study. *Nicotine & tobacco research : official journal of the Society for Research on Nicotine and Tobacco*, ntad234. Advance online publication. <https://doi.org/10.1093/ntr/ntad234>
- Tolonen, H., Reinikainen, J., **Zhou, Z.**, Härkänen, T., Männistö, S., Jousilahti, P., Paalanen, L., Lundqvist, A., & Laatikainen, T. (2023). Development of non-communicable disease risk factors in Finland: projections up to 2040. *Scandinavian journal of public health*, 51(8), 1231–1238. <https://doi.org/10.1177/14034948221110025>

## SKILLS

---

- Statistical software & programming: R (advanced), SPSS (advanced), Python (intermediate), Stata (intermediate).
- Methods: Regression modelling (incl. GLMs), survival analysis, longitudinal analysis, multivariate analysis, multiple imputation, simulation, Bayesian statistics (coursework).

## LANGUAGES

---

- English: C1 (professional).
- Swedish: B1 (SFI D).
- Finnish: basic (currently learning).
- Mandarin and Cantonese: native.

## REFERENCES

---

- Dr. Jaakko Reinikainen (Senior Researcher), Email: [Jaakko.Reinikainen@thl.fi](mailto:Jaakko.Reinikainen@thl.fi)
- Dr. Haoxuan Sa (Postdoctoral Researcher), Email: [haoxuan.sa@helsinki.fi](mailto:haoxuan.sa@helsinki.fi)
- Prof. Matti Pirinen (Professor), Email: [matti.pirinen@helsinki.fi](mailto:matti.pirinen@helsinki.fi)